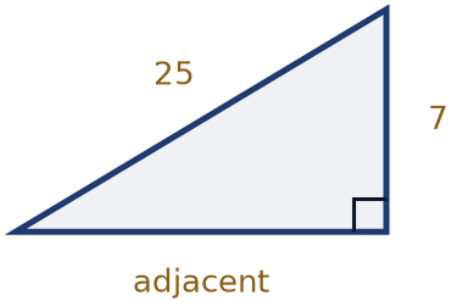
Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos\theta$ and $\tan\theta$.



2 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin\theta$ and the hypotenuse.

3 [Foundational] • Target: Exact values and the unit circle

State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.

4 [Foundational] • Target: Graphs of trigonometric functions

State the amplitude and period of $y = 4\sin(2x)$.

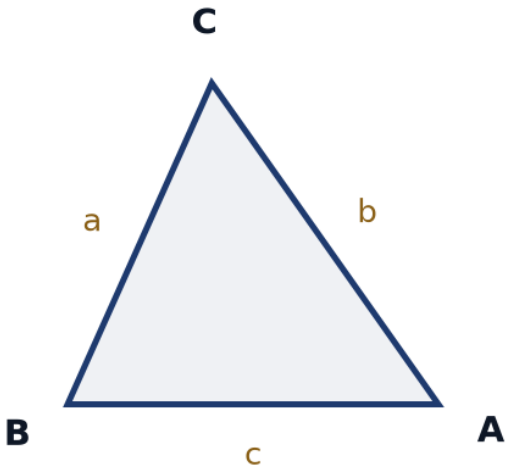
5 [Foundational] • Target: Trigonometric identities

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

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6 [Foundational] • Target: Law of Sines and Law of Cosines

In triangle ABC, $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.



7 [Foundational] • Target: Law of Sines and Law of Cosines

In triangle ABC, angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b, to the nearest tenth, using the Law of Sines.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle ABC, $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.

9 [Intermediate] • Target: Trigonometric identities for compound angles

Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.

10 [Intermediate] • Target: A well-designed word problem: the height of a lighthouse

This question has two parts. A ship observes a lighthouse at an angle of elevation of 18°. After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30°. (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h, to the nearest meter.

11 [Intermediate] • Target: Graphs of trigonometric functions

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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12 **[Intermediate]** • **Target: Trigonometric identities**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 **[Intermediate]** • **Target: Solving trigonometric equations**

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

14 **[Intermediate]** • **Target: Law of Sines and Law of Cosines**

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Trigonometric identities for compound angles** • **Cross-Lesson: 19.01 Normal Distribution**

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

16 **[Challenging]** • **Target: A well-designed word problem: the height of a lighthouse** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

17 **[Challenging]** • **Target: Right-triangle trigonometry** • **Cross-Lesson: 3.01 Graphs & Transformations**

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

18 **[Challenging]** • **Target: Exact values and the unit circle** • **Cross-Lesson: 4.01 Coordinate Geometry**

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 90°$.

19 **[Challenging]** • **Target: Graphs of trigonometric functions** • **Cross-Lesson: 6.01 Differentiation & Integration**

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

20 **[Challenging]** • **Target: Trigonometric identities** • **Cross-Lesson: 7.01 Statistics & Probability**

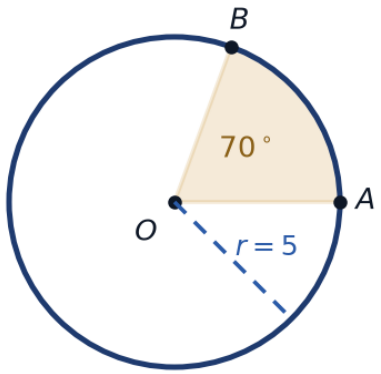
Express $\frac{3x + 1}{(x + 1)(x - 2)}$ in partial fractions.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Right-triangle trigonometry

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

22 [Foundational] • Target: Exact values and the unit circle

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

23 [Foundational] • Target: Graphs of trigonometric functions

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

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24 [Foundational] • Target: Trigonometric identities

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

25 [Foundational] • Target: Solving trigonometric equations

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

26 [Foundational] • Target: Law of Sines and Law of Cosines

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

27 [Foundational] • Target: Trigonometric identities for compound angles

In triangle ABC, the lengths of sides a, b, and c are opposite to angles A, B, and C respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side c.

a. $\sqrt{79}$

b. 9

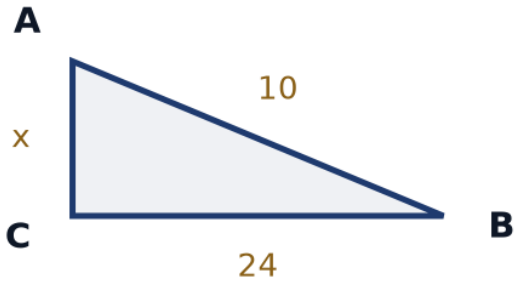
c. $\sqrt{91}$

d. 10

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28 [Foundational] • Target: A well-designed word problem: the height of a lighthouse

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse AB.



a. 25

b. 26

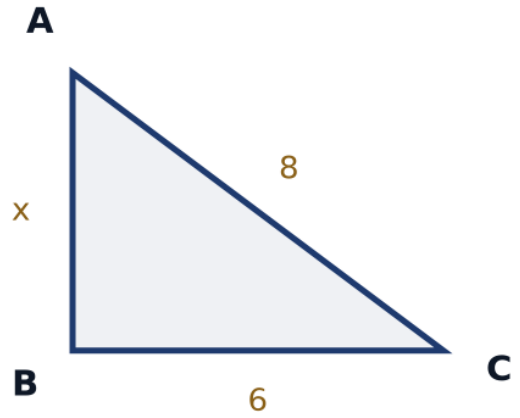
c. 27

d. 28

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Right-triangle trigonometry

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?



a. 10 units

b. 14 units

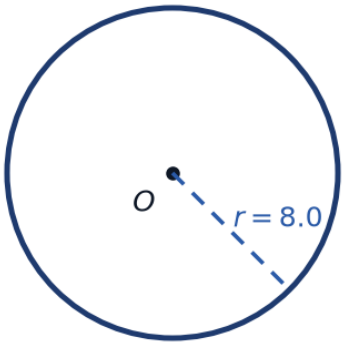
c. 48 units

d. 20 units

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30 [Intermediate] • Target: Exact values and the unit circle

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?



a. $28.8\ \text{cm}^2$

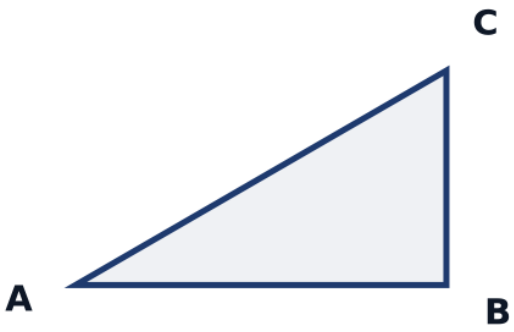
b. $57.6\ \text{cm}^2$

c. $33.8\ \text{cm}^2$

d. $7.2\ \text{cm}^2$

31 [Intermediate] • Target: Graphs of trigonometric functions

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 10$. Find side b (Law of Sines).



a. 10

b. 22

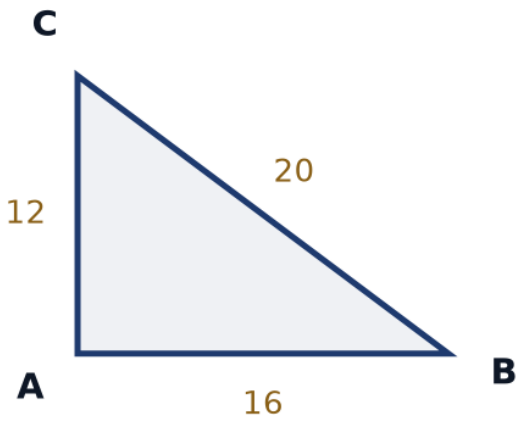
c. 5

d. 20

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32 [Intermediate] • Target: Trigonometric identities

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.



a. $\frac{4}{3}$

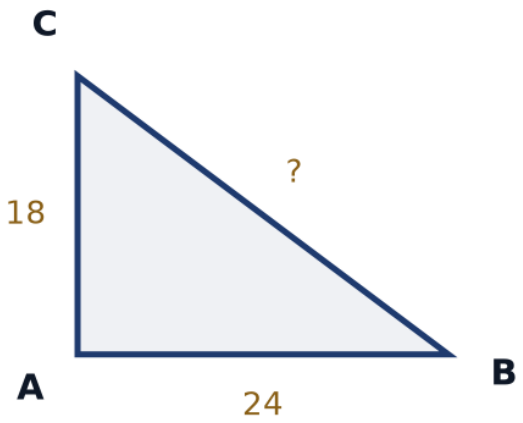
b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$

33 [Intermediate] • Target: Solving trigonometric equations

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.



a. 32

b. 34

c. 33

d. 30

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34 [Intermediate] • Target: Law of Sines and Law of Cosines

The function f is defined on the open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by $f(x) = \frac{2-\sin x}{\cos x} + \frac{\cos x}{2-\sin x}$. Determine the exact minimum value of $f(x)$.

a. $\frac{4\sqrt{3}}{3}$

b. 2

c. $\frac{5}{2}$

d. $\frac{3\sqrt{3}}{2}$

35 [Intermediate] • Target: Trigonometric identities for compound angles

In triangle ABC, the side lengths are $AB = (x+2)$ cm, $BC = (3x-1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.

a. $10\sqrt{3}$ cm^2

b. $5\sqrt{3}$ cm^2

c. $20\sqrt{3}$ cm^2

d. 10 cm^2

36 [Intermediate] • Target: A well-designed word problem: the height of a lighthouse

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

37 [Intermediate] • Target: Right-triangle trigonometry

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

a. 14 cm

b. $2\sqrt{19}$ cm

c. $2\sqrt{34}$ cm

d. 8 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Exact values and the unit circle • Cross-Lesson: 4.01 Coordinate Geometry

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

a. $(24 + 4\pi)$ cm

b. $(12 + 4\pi)$ cm

c. 4π cm

d. $(24 + 24\pi)$ cm

39 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 6.01 Differentiation & Integration

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

a. 6

b. 8

c. 12

d. 15

40 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 7.01 Statistics & Probability

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

a. 5

b. 6

c. 7

d. 8

41 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 8.01 Functions: Composite & Inverse

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. 41.4°

b. 46.4°

c. 138.6°

d. 36.4°

42 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 9.01 Circular Measure

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. 44.4°

d. 39.4°

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43 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 1.01 Indices, Surds & Quadratics

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

44 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 10.01 Series & Binomial Theorem

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120^\circ$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. 14.8

c. 15.8

d. 12.2

45 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 11.01 Kinematics

In triangle ABC, $a = 5$, $b = 10$, and $C = 120^\circ$. Find the exact length of c.

a. $5\sqrt{7}$

b. $\sqrt{179}$

c. 15

d. $3\sqrt{19}$

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